

Proposal Evaluation Form



EUROPEAN COMMISSION

Horizon Europe Framework Programme (HORIZON)

Evaluation Summary
Report

Call: HORIZON-MSCA-2025-PF
Type of action: HORIZON-TMA-MSCA-PF-EF
Proposal number: 101285931
Proposal acronym: GROCOMO-CFT
Duration (months): 24
Proposal title: GROupoid COvariance and Multilocal Observables in Classical Field Theories on curved backgrounds
Activity: EF-MAT

N.	Proposer name	Country	Total eligible costs	%	Grant Requested	%
1	UNIVERSITE LYON 1 CLAUDE BERNARD	FR	242,260.56	100.00%	242,260.56	100.00%
Total:			242,260.56		242,260.56	

Abstract:

The GROCOMO-CFT project (GROupoid COvariance and Multilocal Observables in Classical Field Theories on curved backgrounds) seeks to create a geometric framework for classical field theories on curved spacetimes. It introduces Lie groupoids to model local, observer-dependent symmetries and extends multisymplectic geometry to multilocal observables, addressing the limitations of the Standard Model's global Poincaré symmetry, which applies only to flat spacetime.

The first goal is to construct and analyze the Poincaré groupoid, along with its Spin extension for fermionic matter. This will clarify the connection between the frame groupoid (as jet prolongations of the pair groupoid) and the covariance properties of observer-dependent charts, generalizing global Poincaré invariance in curved Lorentzian manifolds.

The second goal involves developing a groupoid gauge theory of gravitation, defining connections on the Lie algebroid of the Poincaré groupoid, and comparing notions of curvature and torsion with the tetrad and spin-connection formalism. This framework aims to provide tools for addressing Einstein–Cartan and teleparallel regimes.

The third goal is to extend multisymplectic geometry’s structures to multilocal observables, defining L-infinity algebras of observables that are equivariant under groupoid symmetries and creating homotopy comomentum maps. Case studies on scalar fields and selected Standard Model particles on fixed curved spacetimes will evaluate this framework.

By merging differential geometry, higher algebra, and field theory, GROCOMO-CFT aspires to establish a covariant formalism that extends Noether’s theorem and coherently treats multilocal observables, potentially integrating gravity and matter fields within a groupoid-based context.

Evaluation Summary Report

Evaluation Result

Total score: 85.40 % (Threshold: 70 /100.00)

Criterion 1 - Excellence

Score: 4.40 (Threshold: 0 / 5.00 , Weight: 50.00%)

- Quality and pertinence of the project’s research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)
- Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices)

- **Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host**
- **Quality and appropriateness of the researcher's professional experience, competences and skills**

Quality and pertinence of the project's research and innovation objectives:

Excellent

Extent to which the research and innovation objectives are achievable, measurable and verifiable:

Good

Level of ambition of the research and innovation objectives and advancement beyond the state of the art:

Excellent

Soundness of the proposed methodology, including the concepts, models, and assumptions underpinning the project:

Good

Identification of important methodological challenges and proposed measures to tackle them:

Good

Integration of interdisciplinary expertise and methods to pursue the project's objectives:

Excellent

Consideration of the gender dimension and other diversity aspects in the proposal's research and innovation content:

Not applicable

Implementation of appropriate open science practices in the proposed methodology:

Excellent

Quality of the supervision:

Good

Quality of the planned training activities for the researcher:

Good

Overall effectiveness of the transfer of knowledge between the researcher, the host organisation, and, if applicable, the outgoing phase host:

Excellent

Rationale and added value of the non-academic placement:

Not applicable

Rationale and added value of the secondment(s):

Not applicable

Quality of the researcher's professional experience, competences and skills:

Excellent

Appropriateness of the researcher's professional experience in relation to the proposed research project:

Excellent

Comments:

The proposal does not clearly demonstrate that its aims are fully achievable. The absence of a convincing methodology results in an insufficiently clear identification of the key challenges. Additionally, the proposal does not adequately describe the supervisor's experience in post-doctoral mentoring, and the planned training activities are not detailed enough.

Criterion 2 - Impact

Score: 3.70 (Threshold: 0 / 5.00 , Weight: 30.00%)

- Credibility of the measures to enhance the career perspectives and employability of researchers and contribution to their skills development.
- Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities.
- The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts.

Credibility of the measures to enhance the career perspectives and employability of the researcher (including the contribution of their skills development):

Very good

Suitability and quality of the planned dissemination and exploitation measures:

Very good

Suitability and quality of the planned communication and public engagement activities:

Good

Strategy for intellectual property management and protection:

Not applicable

Expected scientific impact beyond the immediate scope and duration of the project:

Fair

Expected economical/ technological and/or societal impacts of the project beyond its immediate scope and duration:

Not applicable

Comments:

The proposal does not provide full details on the specific activities aimed at enhancing the researcher's career prospects. The target audiences for both dissemination and communication activities are insufficiently described. Furthermore, the proposal does not convincingly explain the potential impact and relevance of the project on the scientific community beyond its duration.

Criterion 3 - Implementation

Score: 4.80 (Threshold: 0 / 5.00 , Weight: 20.00%)

- Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages.
- Quality and capacity of the host institutions and participating organisations, including hosting arrangements.

Quality and effectiveness of the work plan, including deliverables and milestones:

Excellent

Appropriateness of effort and timing assigned to each work package:

Excellent

Completeness of the mandatory Gantt Chart and its consistency with the overall work plan:

Very good

Consideration of risks and adequacy of contingency measures:

Good

Quality of hosting arrangements, including team integration and support services:

Excellent

Quality and capacity of all participating organisations, including their infrastructure, logistics, and facilities:

Excellent

Comments:

The Gantt chart does not clearly identify when the project outcome will be delivered at conferences or seminars. The mitigation strategy described in the proposal is not fully convincing and the administrative risks are not addressed in enough detail.

Scope of the application

Status: Yes

Comments (in case the proposal is out of scope)

Not provided

Exceptional funding

A third country participant/international organisation not listed in [the General Annex to the Main Work Programme](#) may exceptionally receive funding if their participation is essential for carrying out the project (for instance due to outstanding expertise, access to unique know-how, access to research infrastructure, access to particular geographical environments, possibility to involve key partners in emerging markets, access to data, etc.). (For more information, see the [HE programme guide](#))

Please list the concerned applicants and requested grant amount and explain the reasons why.

Based on the information provided, the following participants should receive exceptional funding:

Not provided

Based on the information provided, the following participants should NOT receive exceptional funding:

Not provided

Use of human embryonic stem cells (hESC)

Status: No

If YES, please state whether the use of hESC is, or is not, in your opinion, necessary to achieve the scientific objectives of the proposal and the reasons why. Alternatively, please state if it cannot be assessed whether the use of hESC is necessary or not, because of a lack of information.

Not provided

Use of human embryos

Status: No

If YES, please explain how the human embryos will be used in the project.

Not provided

Activities excluded from funding

Status: **No**

If YES, please explain.

Not provided

Exclusive focus on civil applications

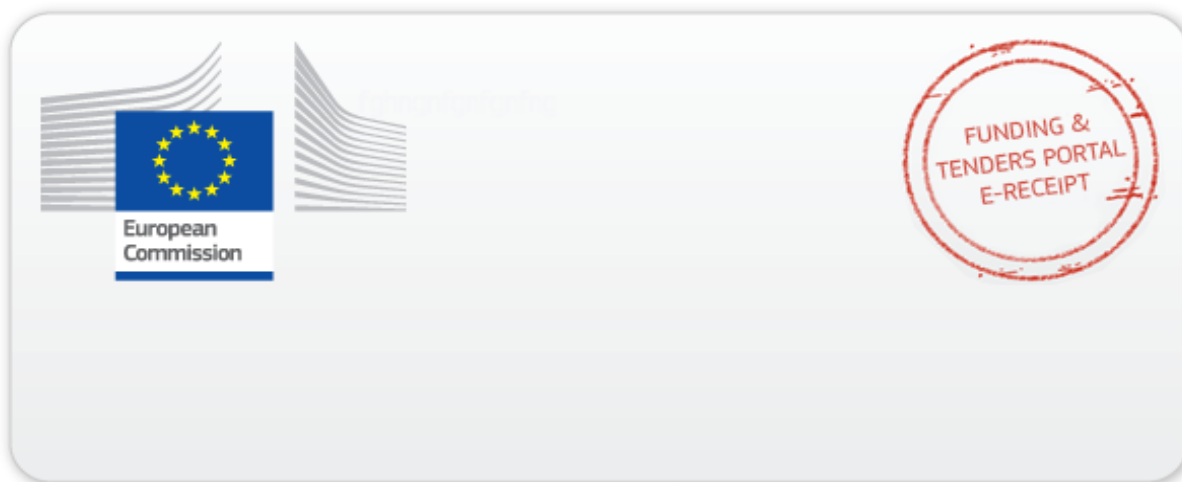
Status: **Yes**

If NO, please explain.

Not provided

Overall comments

Not provided



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